

Удивительный мир VSR метрик



Таксономия метрик

1. Level

temporal

frame-level

subjective

2. Computation logic

pixel-based

DL

3. Evaluation intent

fidelity

perceptual
quality

realism

How much does the output
look like the pixel-perfect
Ground Truth?

How "sharp" or "pleasing"
does it look to a human?

Does the added detail
look like a real video or AI
noise?

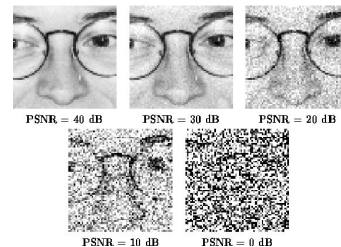
↑ PSNR (Peak Signal-to-Noise Ratio)

frame-level

pixel-based

fidelity

+ full-reference



How does it work?

Given a noise-free $m \times n$ monochrome image I and its noisy approximation K , MSE is defined as $MSE = \frac{1}{m \cdot n} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} [I(i, j) - K(i, j)]^2$

The PSNR is defined as

$$PSNR = 10 \cdot \log_{10} \left(\frac{MAX_I^2}{MSE} \right)$$

Here, MAX_I is the maximum possible pixel value of the image. When the pixels are represented using 8 bits per sample, this is 255.

For RGB color images, with three values per pixel, the definition of PSNR is the same except that the MSE is the sum over all squared value differences (now for each color, i.e. three times as many differences as in a monochrome image) divided by image size and by three.

↑ SSIM (Structural Similarity Index)

frame-level

pixel-based

fidelity

+ full-reference

How does it work?

Image degradation = perceived change in structural information

Structural information is the idea that the pixels have strong inter-dependencies especially when they are spatially close.

Two key perceptual phenomena: **luminance masking**, where distortions are less visible in bright regions, and **contrast masking**, where distortions are hidden by significant texture or image activity.

The SSIM index is calculated between two windows of pixel values x and y of common size, from corresponding locations in two images. These SSIM values can be aggregated across the full images by averaging or other variations.

$$\text{SSIM}(x, y) = l(x, y)^\alpha \cdot c(x, y)^\beta \cdot s(x, y)^\gamma$$

$$l(x, y) = \frac{2\mu_x\mu_y + c_1}{\mu_x^2 + \mu_y^2 + c_1}$$

$$c(x, y) = \frac{2\sigma_x\sigma_y + c_2}{\sigma_x^2 + \sigma_y^2 + c_2}$$

$$s(x, y) = \frac{\sigma_{xy} + c_3}{\sigma_x\sigma_y + c_3}$$

↑ SSIM (Structural Similarity Index)

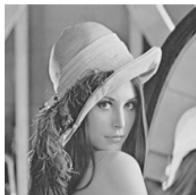
frame-level

pixel-based

fidelity

+ full-reference

MSE isn't able to capture blurring



GT



Denoised image



Denoised image



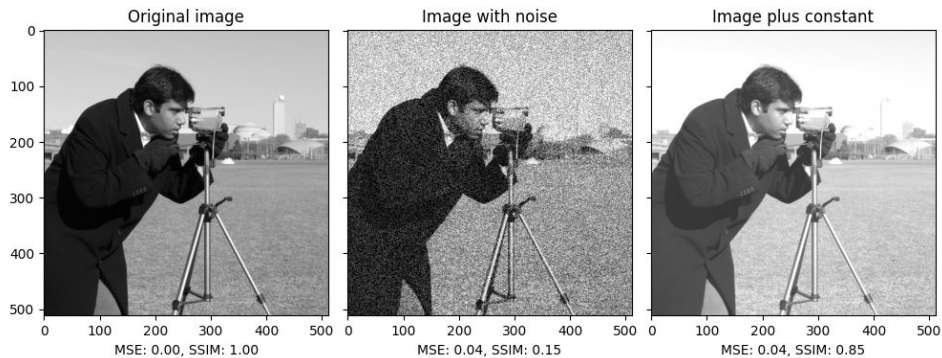
Denoised image



Denoised image



Denoised image



↑ MS-SSIM (MultiScale Structural Similarity Index)

frame-level

pixel-based

fidelity

+ full-reference

How does it work?

MS-SSIM (Multi-Scale Structural Similarity) addresses a major limitation of the original SSIM: human perception of image quality depends heavily on the viewing distance and the resolution of the display. A high-frequency artifact might be jarringly obvious up close but completely invisible if you step back or view the image on a smaller screen.

$$MS-SSIM(x, y) = [l_M(x, y)]^{\alpha_M} \cdot \prod_{j=1}^M [c_j(x, y)]^{\beta_j} [s_j(x, y)]^{\gamma_j}$$

Think of M as the "farthest viewing distance." The process works through a series of stages:

Scale 1: The original, full-resolution image.

Scale 2: The image is downsampled by a factor of 2 (blurring and shrinking).

Scale 3 to M : This downsampling continues iteratively until the image reaches the final scale, M .

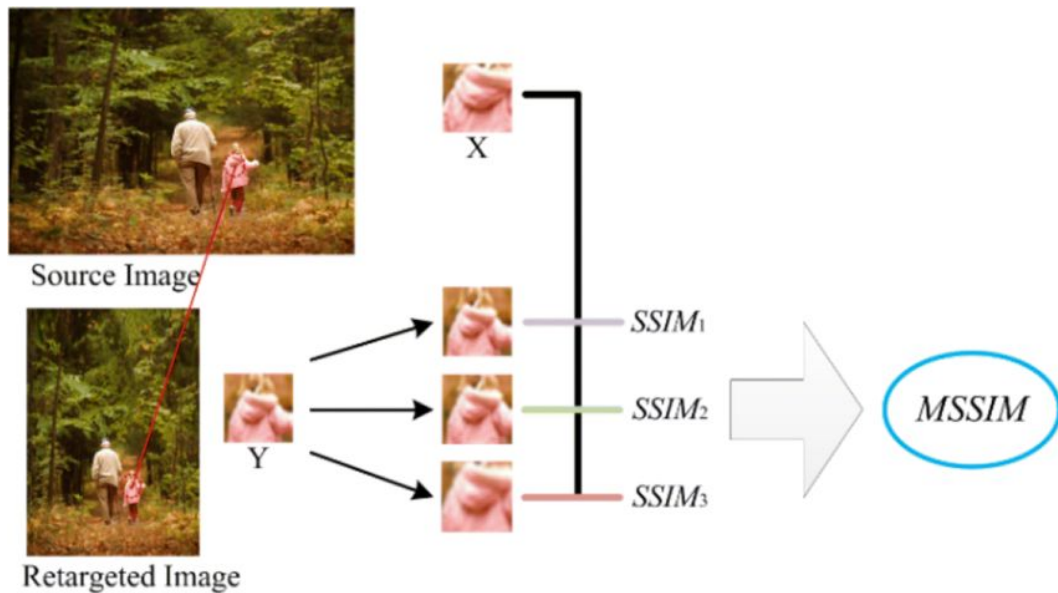
↑ MS-SSIM (MultiScale Structural Similarity Index)

frame-level

pixel-based

fidelity

+ full-reference



↑ IFC (Information Fidelity Criterion)

frame-level

pixel-based

fidelity

+ full-reference

How does it work?

Shifts the focus from structural similarity to **Information Theory**.

It treats the process of image distortion as a loss of information across a communication channel.

IFC is based on the idea that "natural" images (like a photo of a forest or a face) have specific statistical properties that distinguish them from random noise or artificial distortions. These properties are called **Natural Scene Statistics (NSS)**.

When an image is super-resolved, the SR algorithm acts as a channel that potentially "leaks" or destroys the original information. IFC quantifies how much of the original, high-fidelity information can still be extracted from the distorted version.

Using Shannon's Information Theory, the metric calculates the **Mutual Information** between the source (reference) and the output (distorted image).

↓ LPIPS (Learned Perceptual Image Patch Similarity)

frame-level

DL

perceptual
quality

+ full-reference

How does it work?

The Intuition: "Deep Features" as a Proxy for Human Vision

Researchers discovered that deep CNNs, when trained on massive datasets for image recognition, naturally develop filters that mimic the human visual system. LPIPS leverages these pre-trained "feature maps" to compare two images.

- 1) **Feature Extraction**: Both the reference image and the SR image are passed through a pre-trained network.
- 2) **Activation Collection**: The activations (feature maps) are extracted from several layers of the network.
- 3) **Distance Calculation**: The difference between these feature maps is calculated, typically using an L2 distance, and then weighted.
- 4) **Learning from Humans**: These weights are not random; they were specifically trained on a dataset called BAPPS (Berkeley-Adobe Perceptual Patch Similarity), where humans voted on which of two distorted images looked more like the original.

Gold standard for GAN-based (Generative Adversarial Network) super-resolution. **The "Texture" Problem**: PSNR and SSIM often penalize GANs for generating realistic textures (like blades of grass or pores on skin) if those textures don't align perfectly with the ground truth pixels.

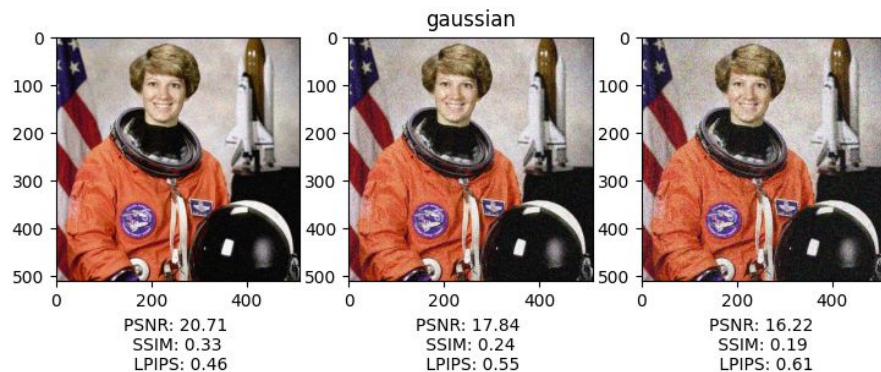
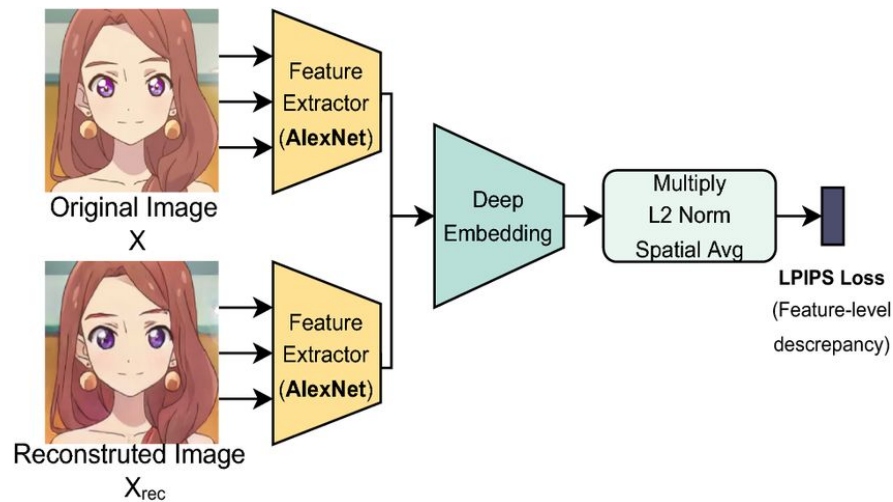
↓ LPIPS (Learned Perceptual Image Patch Similarity)

frame-level

DL

perceptual
quality

+ full-reference



↑ VMAF (Video Multi-Method Assessment Fusion)

frame-level

DL

perceptual
quality

+ full-reference

How does it work?

From Netflix with love

VMAF is an **ensemble metric** — it combines multiple quality scores into one using a machine learning model.

It uses a **Support Vector Machine (SVM) regressor** that has been trained on a massive dataset of human subjective scores (Mean Opinion Scores).

Combines these three primary features:

- 1) **VIF (Visual Information Fidelity)**: measures the loss of information between the reference and the distorted image across multiple scales
- 2) **DLM (Detail Loss Metric)**: measures the loss of details that are visible to the human eye, as well as the introduction of "additive" impairments
- 3) **TI (Temporal Information)**: it looks at the motion between frames

The industry standard for evaluating how SR affects the "Quality of Experience" (QoE) for end-users

↓ FID (Fréchet Inception Distance)

frame-level

DL

realism

+ full-reference

How does it work?

"Does the statistical distribution of my generated images match the distribution of real-world images?"

- 1) **Feature Extraction:** Both the set of real images and the set of generated images are passed through a pre-trained Inception-v3 network. The output isn't the final classification (e.g., "cat"), but the activations from the last pooling layer.
- 2) **Modeling the Distribution:** We assume these features follow a **Multivariate Gaussian Distribution**. We calculate the mean (μ) and covariance (Σ) for both the real (r) and generated (g) sets.
- 3) **The Fréchet Distance:** We calculate the "distance" between these two Gaussian distributions. Mathematically, this is the Wasserstein-2 distance:

$$d^2 = \|\mu_r - \mu_g\|^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2})$$

- 1) **Detects fake and plastic features** (feature distributions will not match real images)
- 2) **Detects model collapse** (If an SR model starts producing the same texture for every tree or every building, the covariance of the generated set will be low)

↓ FVD (Fréchet Video Distance)

frame-level

DL

realism

+ full-reference

How does it work?

The Core Difference: 2D vs. 3D Features

Uses **I3D** (Inflated 3D ConvNet). This network uses 3D convolutions that process multiple frames at once.

Recent research has shown that FVD is actually heavily biased toward image quality. Because the I3D backbone was originally trained for action recognition, it sometimes prioritizes the "objects" in the frame over the "smoothness" of the motion.

$$FVD = \|\mu_{real} - \mu_{gen}\|^2 + Tr(\Sigma_{real} + \Sigma_{gen} - 2(\Sigma_{real}\Sigma_{gen})^{1/2})$$

- 1) **Detects fake and plastic features** (feature distributions will not match real images)
- 2) **Detects model collapse** (If an SR model starts producing the same texture for every tree or every building, the covariance of the generated set will be low)

↑ ERQA (Edge Restoration Quality Assessment)

frame-level

DL

perceptual
quality

+ full-reference

How does it work?

ERQA focuses exclusively on edges

ERQA provides a quantitative measure of how well the edge structures have been reconstructed compared to the Ground Truth (GT).

- 1) Edge Detection: with Canny edge detector or a Sobel filter.
- 2) Edge Thinning and Alignment: the edges are "thinned" to a single pixel width (a blurry SR image might overlap more with the Ground Truth). The metric then aligns the edge maps to account for any sub-pixel shifts that might have occurred during the SR process.
- 3) Edge Recall: how many of the "real" edges from the Ground Truth were successfully recovered by the SR model (low = model is smoothing over the details)
- 4) Edge Precision: how many of the edges in the SR image actually exist (low = model hallucinates fake edges or creates artifacts)
- 5) F-score

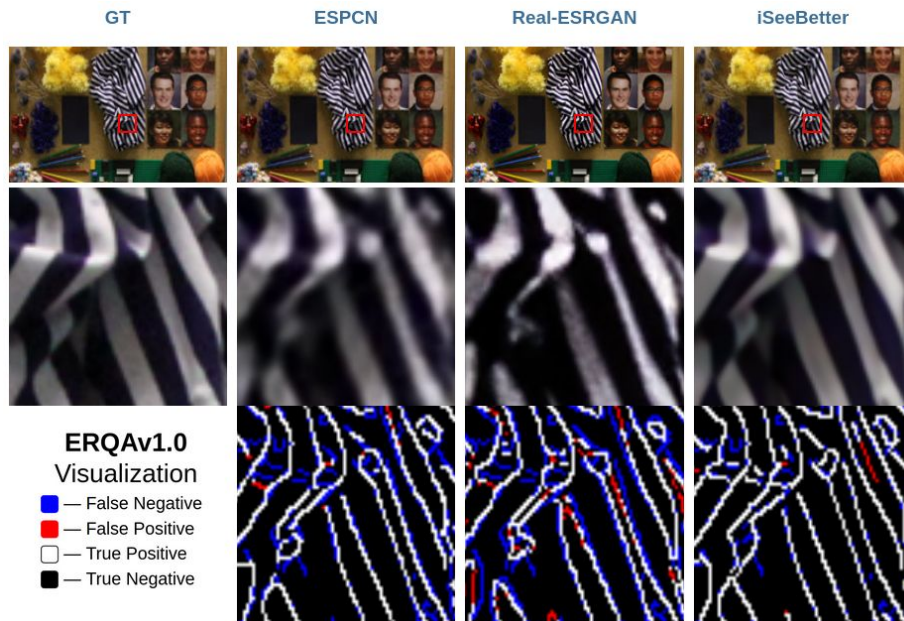
↑ ERQA (Edge Restoration Quality Assessment)

frame-level

pixel-based

fidelity

+ full-reference



↓ CER (Color Error Rate)

frame-level

pixel-based

fidelity

+ full-reference

How does it work?

Isolate and quantify color distortions

In SR, models often focus so much on sharpening edges that they accidentally "bleed" colors or shift the white balance of small objects.

- 1) **Color Space Conversion**: Both the Ground Truth (GT) and the SR image are converted from RGB to a space like **CIELAB** (separates color from brightness)
- 2) **Euclidean Distance**: For every pixel, the metric calculates the distance between the color vectors of the GT and SR images.
- 3) **Thresholding**: In some implementations, a "rate" is calculated by determining what percentage of pixels exceed a specific ΔE threshold (the point where a human would notice the color is "off").

↓ tOF (Temporal Optical Flow Error)

temporal

pixel-based

fidelity

+ full-reference

How does it work?

Looks at the "path" pixels take from one frame to the next.

$$tOF = \sqrt{(u_{ref} - u_{sr})^2 + (v_{ref} - v_{sr})^2}$$

If the SR model produces sharp frames but slightly shifts the car's position inconsistently in each frame, the video will "stutter" or "jitter." tOF quantifies the jitter by comparing the **optical flow** (the velocity vectors of pixels) of the SR video against the Ground Truth.

1. **Optical Flow Estimation:** an external algorithm is used to calculate the motion vectors for both the Reference (V_{ref}) and the Super-Resolved (V_{sr}) video. For every pixel, this provides a 2D vector (u, v) representing its horizontal and vertical movement.
2. **Calculating the Error:** the metric calculates the difference between the motion vectors. This is typically done using the Euclidean distance
3. **Aggregation:** These errors are calculated for every pixel in every frame and then averaged across the entire video sequence.

↓ tOF (Temporal Optical Flow Error)

temporal

pixel-based

fidelity

+ full-reference



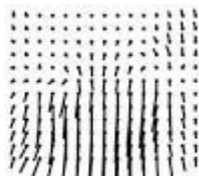
(a)



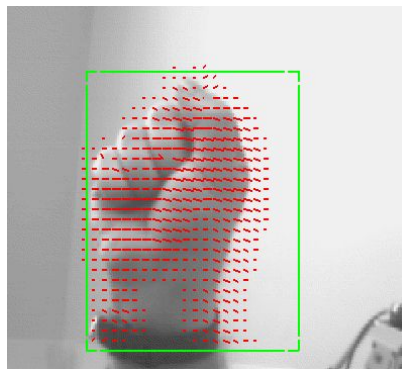
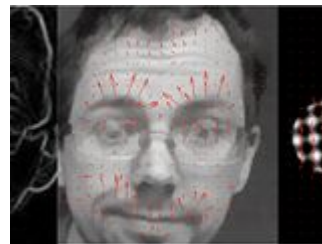
(b)



(c)



(d)



↓ tLP (Temporal LPIPS)

$$tLP = |LPIPS(SR_t, SR_{t+1}) - LPIPS(GT_t, GT_{t+1})|$$

temporal

DL

perceptual
quality

+ full-reference

How does it work?

The Intuition: Perceptual Flickering

In Video Super-Resolution, a model might produce frames that look great individually (low LPIPS) but "flicker" when played back. This happens because the deep learning model might reconstruct a texture slightly differently in Frame 1 than it does in Frame 2.

It uses the same deep feature extraction as **LPIPS** but applies it across the time dimension.

- 1) **Feature Change in GT**: Calculate the **LPIPS** distance between two consecutive frames in the GT video (GT_t and GT_{t+1}). This represents the "natural" motion or change.
- 2) **Feature Change in SR**: Calculate the **LPIPS** distance between the same two consecutive frames in the SR video (SR_t and SR_{t+1}).
- 3) **Distance Comparison**: tLP is the absolute difference between these two perceptual changes.

↑FasterVQA (Faster Video Quality Assessment)

temporal

DL

perceptual
quality

+ no reference

How does it work?

Solve a massive problem in video evaluation: computational cost.

The breakthrough of FasterVQA is a technique called **Grid Mini-patch Sampling (GMS)**

You usually have to choose between Resize (which blurs fine details) or Crop (which ignores most of the image). *GMS* avoids this trade-off.

- **Fragmenting**: *GMS* divides a high-resolution video frame into a grid. From each grid cell, it samples a tiny "mini-patch" (e.g., 32×32 pixels).
- **Splicing**: It then stitches these tiny patches together into a single, smaller "fragmented" image.
- **You get a low-resolution input that still contains raw, unscaled pixels**. This allows the model to see fine-grained SR artifacts (like aliasing or noise) while only having to process a fraction of the total data.

↑ FasterVQA (Faster Video Quality Assessment)

temporal

DL

perceptual
quality

+ no reference

How does it work?

Solve a massive problem in video evaluation: computational cost.

Then it uses a specialized architecture called a **Hierarchical Architecture Transformer (HAT)**.

Space-Time Attention: HAT doesn't just look at frames individually. It uses **Temporal Attention** to compare patches across different points in time.

- 1) **Fragment Input**: The sampled fragments are fed into the network. Because the total number of pixels in these fragments is much smaller than the full video, the model runs significantly faster.
- 2) **Feature Extraction**: The Transformer layers look for both spatial distortions (like blur) and temporal distortions (like "shimmering" or jitter).
- 3) **Quality Regression**: The model outputs a final quality score that predicts how a human would rate the video.

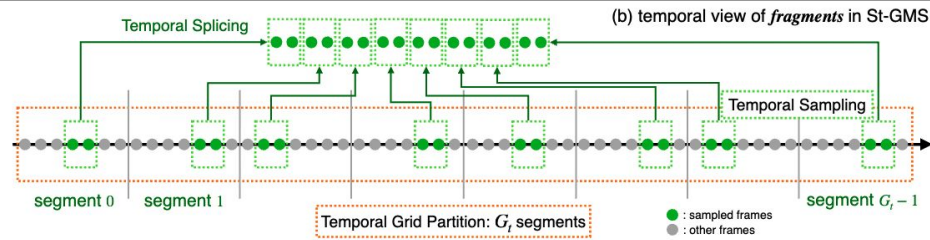
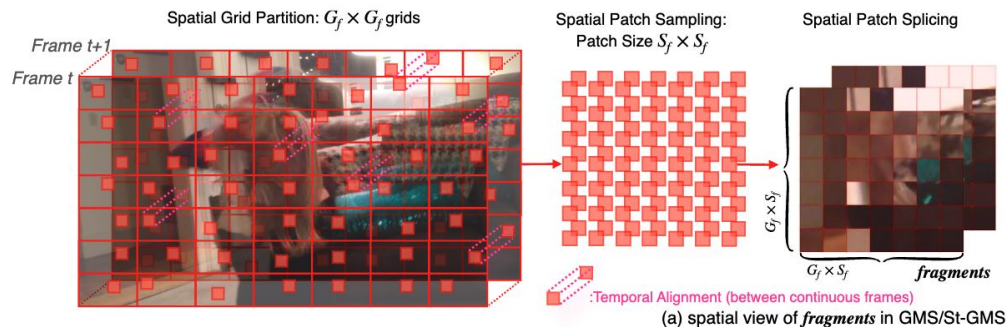
↑ FasterVQA (Faster Video Quality Assessment)

temporal

DL

perceptual
quality

+ no reference



↑ DOVER (Disentangled Objective Video Quality Evaluator)

temporal

DL

perceptual
quality

+ no reference

How does it work?

The Core Philosophy: Aesthetic vs. Technical.

1. The Aesthetic Branch (Global View)

This branch takes the entire frame, resizes it, and looks at the big picture. It uses a Swin Transformer (a powerful deep learning model) to judge how visually pleasing the content is. It doesn't care about a few missing pixels; it cares about the overall look.

2. The Technical Branch (Local View)

GMS (Grid Mini-patch Sampling) comes in. Instead of looking at the whole frame, it samples tiny, raw-resolution patches. This allows it to see "micro-distortions" like pixel-level artifacts that would be lost if the image were resized.

↑ DOVER (Disentangled Objective Video Quality Evaluator)

temporal

DL

perceptual
quality

+ no reference

How does it work?

The Core Philosophy: Aesthetic vs. Technical.

SR model produces a sharp image that looks artificial → the Technical score might be **high**, but the Aesthetics core will be **low**.

SR model produces a "beautiful" but blurry image → the Aesthetic score might be **high**, but the Technical score will be **low**.

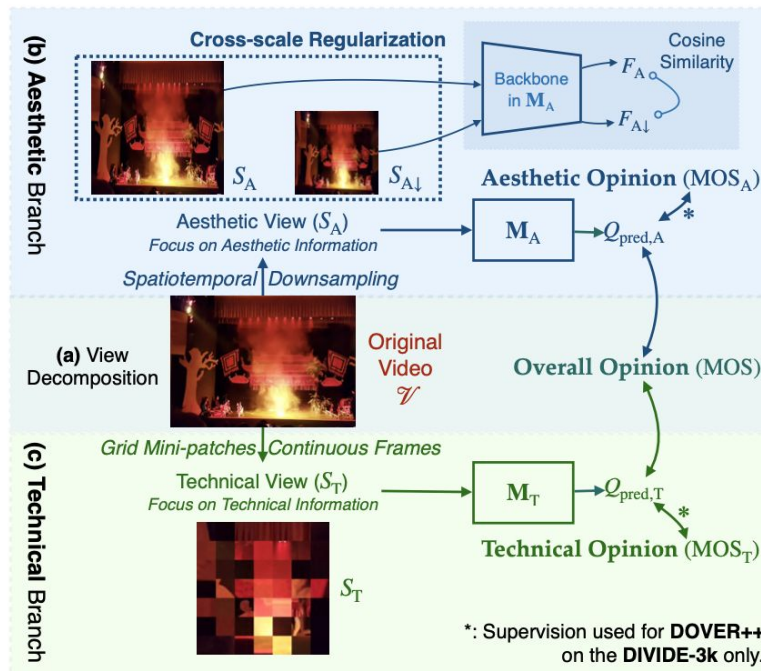
↑ DOVER (Disentangled Objective Video Quality Evaluator)

temporal

DL

perceptual
quality

+ no reference



↑ DOVER (Disentangled Objective Video Quality Evaluator)

temporal

DL

perceptual
quality

+ no reference

How does it work?

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↑ CLIP-IQA

frame-level

DL

perceptual
quality

+ no reference

How does it work?

Любимая метрика Максима

CLIP-IQA doesn't need a Ground Truth image, it places the image into a space where it can be compared against antonym text pairs (e.g., "Good photo" vs. "Bad photo").

1. The metric generates embeddings for a pair of prompts:
 - **Positive Anchor:** "Good image," "Sharp photo," or "High quality."
 - **Negative Anchor:** "Bad image," "Blurry photo," or "Low quality."
2. The Super-Resolved image is passed through the CLIP image encoder to get its vector representation.
3. The metric calculates how close the image vector is to the "Good" prompt versus the "Bad" prompt using a softmax function.

↑ CLIP-IQA

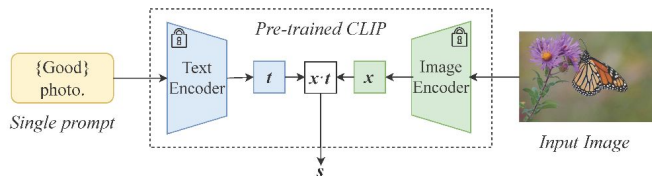
frame-level

pixel-based

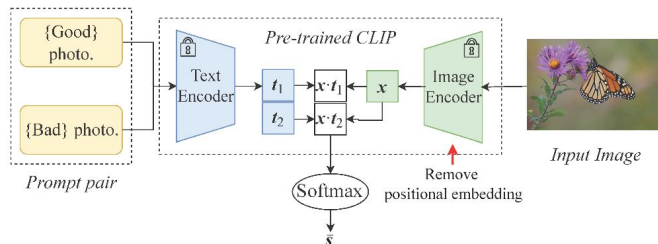
fidelity

+ no reference

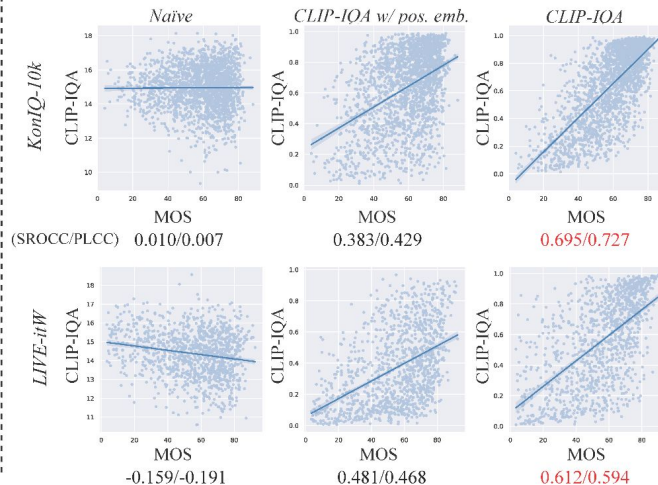
(a) Naïve way to adopt CLIP for visual perception



(b) Framework for the proposed CLIP-IQA



(c) CLIP-IQA score vs. MOS score under different strategies



↑ MOS / CMOS (Mean Opinion Score / Comparative MOS)

subjective

+ no reference

How does it work?

The Gold Standard.

MOS is prone to "calibration drift." One person's "4" might be another person's "3" because they haven't seen the Ground Truth for comparison. The Scale is usually the standard (1 to 5)

CMOS is a **Side-by-Side** metric.

In Super-Resolution research, this is often used to compare a new AI model against an older one.

- The Scale: Usually ranges from -3 to +3: +3 is **Much Better** and -3 is **Much Worse**
- The final score is the **CMOS Gap**. If System A has a CMOS of +0.5 compared to System B, it means System A is perceptibly "better" to the average viewer.

ТОП любимых метрик резерчеров

1. PSNR
2. SSIM
3. LPIPS
4. VMAF
5. DOVER

- Exploring Real-Time Super-Resolution: Benchmarking and Fine-Tuning for Streaming Content (accepted to ICLR-2026): subj., PSNR, LPIPS, CLIP-IQA
- MSU Super-Resolution for Video Compression Benchmark: PSNR, MS-SSIM, VMAF, LPIPS, ERQA
- A Survey of Deep Learning Video Super-Resolution (2025): PSNR, SSIM
- DOVE: Efficient One-Step Diffusion Model for Real-World Video Super-Resolution (2025): PSNR, LPIPS, CLIP-IQA, DOVER
- RealisVSR: Detail-enhanced Diffusion for Real-World 4K Video Super-Resolution (2025): PSNR, SSIM, LPIPS, DOVER

спасибо за внимание

**если есть вопросы
загуглите**